# V S KAMALESH

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## EDUCATION

**Chennai Institute of Technology** Chennai, India
Bachelor of Engineering in Electronics and Communication Engineering (ECE) [CGPA: 7.53] 2024 – 2028

## EXPERIENCE

**ULOG3 — Thermal Engineer** **Jan 2025 – Present**

- Conducting R&D on thermal management architectures and battery system design for CubeSat platforms.
- Evaluating methods to improve thermal stability, battery safety, and operational reliability for space-based applications.

**Chennai Rail Metro Limited — Maintenance Technician Intern** **May 2025 – Jun 2025**

- Studied and documented embedded control systems in metro trains including door modules, pantographs, battery systems, and onboard diagnostics.
- Understood telecom infrastructure across metro stations, including fiber based MAN/LAN deployments and integration of real-time passenger information systems.

## TOP PROJECTS

**Model Rocket V1 / Project Apex — High-Power 3D-Printed Rocket & Telemetry Avionics**
**Technologies:** SolidWorks, Fusion 360, OpenRocket, ANSYS (Fluent/CFD), 3D Printing, ESP32, IMU, Barometer, RF Downlink

- Engineered a custom 3D-printed model rocket airframe, nose cone, and recovery bay from the motor up; modeled fin planform aerodynamics and CP/CG stability margins in OpenRocket, verified supersonic drag profiles via ANSYS Fluent CFD.
- Designed and manufactured onboard avionics integrating an IMU and digital barometer for continuous high-rate telemetry logging .

**ENTITY — Adaptive On-Device LLM Runtime for Arm Silicon**
**Technologies:** C++, Kotlin, Arm NEON/v8-A Optimization, llama.cpp, Android Studio, Android NDK, INT8 Quantization

- Engineered an adaptive, fully offline on-device Large Language Model runtime specifically optimized for Arm-based Android processors with custom kernel-level performance enhancements and cache-aligned memory access.
- Achieved 47% faster token generation compared to Arm's official reference runtime on identical hardware; authored an optimization fix successfully accepted and merged upstream into the official llama.cpp repository.

**Ultstat — Native Apple Silicon System Monitor & Battery Charge Limiter**
**Technologies:** Swift, Xcode, Apple Silicon Low-Level APIs (IOKit, SMC, Mach Kernel), SwiftUI, macOS

- Engineered a lightweight, native menu-bar system monitor and battery lifecycle controller for Apple silicon Macs, interfacing directly with IOKit and SMC for CPU/GPU cluster telemetry, thermals, and power consumption.
- Implemented battery charge limiting, sailing, and calibration algorithms to extend hardware lifespan, maintaining an ultra-low footprint of 0.85 ms/s idle CPU consumption without measurable battery drain.

## ACHIEVEMENTS & HONORS

- **BRICS Hi-Tech Future Skills Competition 2026 (Yekaterinburg, Russia):** Officially invited and selected to represent Chennai Institute of Technology / India in the Aerial Robotics skill (November 2026) – Fully Sponsored by the Organization.
- **NIDAR & IROCU:** Shortlisted as National Finalist for NIDAR (National Drone Aerial Robotics) and qualified as national Semifinalist for IROCU (national Robotics Competition).
- **Lam Research Challenge 2025:** Selected as Top 40 Team across India (National Level) in the prestigious semiconductor engineering challenge.
- **Hackathon Wins:** Won 1st Place (₹15,000 cash prize) at AICon'24 (24Hr Hackathon, Sairam College),1st & 3rd Place at Camri Hackathon CIT.
- **AMD Slingshot Hackathon:** Ranked Top 8 Team representing Chennai at the regional level in edge computing and acceleration.
- **Open Source Contribution :** Upstream contributor to llama.cpp for low level Arm architecture optimization

## TECHNICAL SKILLS

- **Programming Languages:** C, C++ (Embedded), Python (Embedded / ML), Swift.
- **AI & Agentic Systems:** AI Harnesses & Agentic Workflows, On-Device LLM Optimization, YOLOv8, OpenCV, PyTorch, INT8 Quantization
- **CAD, Aerospace & Simulation:** SolidWorks, Autodesk Fusion 360, ANSYS (Fluent / FEA), OpenRocket, Gazebo, ROS, QGroundControl, Mission Planner
- **Embedded Systems & Hardware:** Raspberry Pi 5, ESP32, STM32, Arduino Series, Jetson Series, Cube Orange / PX4 Flight Controllers, BMS, KiCad, Atlium.
- **IDEs, Tools & Protocols:** Xcode, Android Studio, MATLAB / Simulink, Linux / WSL, Git, GitHub, Docker, UART, SPI, I2C, PWM, ADC

## CERTIFICATIONS & LANGUAGES

- **Certifications:** Cisco (Introduction to IoT), Microsoft (Career Essentials in Generative AI), LinkedIn Learning (Electronic Foundations), Google & Microsoft Affiliated (Certified Game Developer)
- **Languages:** English (Fluent / Professional Proficiency), Tamil (Native), Hindi (Basic)